

Cite this: *RSC Adv.*, 2014, 4, 17555

The preparation of MnFe_2O_4 decorated flexible graphene wrapped with PANI and its electrochemical performances for hybrid supercapacitors†

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A modest, cost effective and eco-friendly method is employed for the preparation of MnFe_2O_4 nanocube decorated, flexible graphene sheets, followed by polyaniline (PANI) which is wrapped by an *in situ* chemical oxidative polymerization method. The formation of the hybrid composites and their individual constituents are realized through X-ray diffraction (XRD), Fourier transforms infrared spectroscopy (FT-IR) and Raman spectroscopy. The MnFe_2O_4 nanocube decorated flexible graphene and PANI wrapped composite is visualised through transmission electron microscopic (TEM) images. The decorated MnFe_2O_4 particles have sizes in the range of 75–100 nm. The capacitive properties of the ternary composite is investigated and it has the obtained specific capacitance of 338 F g^{-1} at 0.5 mA cm^{-2} in 1 M NaCl aqueous electrolyte, which is 10 times greater than the pristine MnFe_2O_4 (32 F g^{-1}). Similarly, the fabricated hybrid supercapacitor (MGP||AC) provides the specific capacitance of about 51.87 F g^{-1} at 5 mV s^{-1} and its energy density is $10.25 \text{ W h kg}^{-1}$ at 10 mA cm^{-2} . Thus the obtained significant properties of the ternary composite are due to the distinctive characteristics of the individual constituents which leads to enhanced electrochemical properties like reducing the internal resistance and diffusion path length via the synergistic effect.

 Received 16th December 2013
Accepted 12th February 2014

DOI: 10.1039/c3ra47681b

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1. Introduction

In this fast growing world, researchers are under demands to discover new materials and devices that should be eco-friendly, have a high energy and also have a high power density, so that they can replace fossil fuels, the amounts of which are reducing day by day. In this juncture, the presently used energy storage and conversion devices are batteries, conventional capacitors (CCs), fuel cells and supercapacitors. Generally, power density is required for high power applications like hybrid electrical vehicles, computer backup and power breaking systems. Among these, batteries are widely used energy storage devices, where the energy is stored by a Faradaic reaction and has a high energy density compared to CCs and supercapacitors. However, they have the precincts of a low power density, less cyclic life, a high fabrication cost and they contain toxic electrolytes. Supercapacitors are considered as the ideal storage device to overcome the above mentioned difficulties since they have a high power density, a long cycle life, a long shelf life, operate in all weather conditions, are environmental friendly and safer.

However, they have a low energy density when compared with batteries. Hence, people are urged to identify new materials and fabricate a high energy supercapacitor without losing the above mentioned benefits.^{1–5}

Generally, supercapacitors are classified into three types such as (i) pseudocapacitors, (ii) electrical double layer capacitors (EDLCs), and (iii) hybrid supercapacitors based on their charge storage mechanisms. Currently, hybrid supercapacitors received much attention owing to their improved working potential window, high specific capacitance and energy density compared to pseudocapacitors and EDLCs.^{1,2,5} In hybrid supercapacitors, one electrode stores the charge based on a redox reaction and the other one based on an EDLC.^{1–5b} Hence, activated carbons, carbon nanotubes, carbon aerogels and graphene are used as material for one of the active electrode.^{6–9} In this competition, graphene evolves as the novel electrode due to its superior properties of a high surface area, a 2-dimensional structure, a high thermal and mechanical stability, a high electrical conductivity and flexibility. However, the reattachment of graphene is unavoidable during the electrochemical reaction which inhibits the better electrochemical performance and reduces the cyclic stability. In order to avoid this reattachment, metal oxides or a conducting polymer has been wrapped on a graphene matrix.^{10,11} Considering the spinel ferrites, MnFe_2O_4 is environmental friendly, has multiple valence, and is

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† Electronic supplementary information (ESI) available. See DOI: 10.1039/c3ra47681b

used as an anode for Li-ion batteries and drug delivery applications. Even though it has potential applications in diverse fields, the electrical conductivity of MnFe_2O_4 is not appreciable,^{12–15} which limits its applications in supercapacitors. In order to improve the electrochemical performance an attempt has been made to make a composite with electrically conductive carbon based materials. In this line various metal oxide/graphene composites have been used for supercapacitor applications.^{16–19} Similarly, polypyrrole, polythiophene and polyaniline (PANI) have been widely used conducting polymers for supercapacitors. Among these, PANI is a suitable material for a ternary composite owing to its ease of synthesis, good electrical conductivity, and eco-friendliness. However, the commercialization is prohibited due to its low cycling stability.^{20–22}

In this regard, various ternary hybrid composites have been prepared for supercapacitors. Prepared electrodes for supercapacitors including RGO/MoO₃/PANI (553 F g⁻¹ at 1 mV s⁻¹),²³ RGO/Fe₂O₃/PANI (638 F g⁻¹ at 1 mV s⁻¹),²⁴ RGO/TiO₂/RGO (114.5 F g⁻¹ at 5 mV s⁻¹),²⁵ MnO₂/PPy/Carbon nanofiber (705 F g⁻¹ at 2 mV s⁻¹),²⁶ graphene/PANI/MCM-41 (405 F g⁻¹ at 0.8 Ag⁻¹),²⁷ graphene/PPy/CNT (361 F g⁻¹ at 0.2 Ag⁻¹),²⁸ graphene/PANI/CNT (569 F g⁻¹ at 0.1 Ag⁻¹),²⁹ GE/SnO₂/PEDOT (183 F g⁻¹ at 1 mV s⁻¹),^{30a} SWCNT/MnO₂/rice paper (260.2 F g⁻¹ at 0.5 Ag⁻¹),^{30b} CNT/PANI/MnO₂ (330 F g⁻¹ at 5 mV s⁻¹),^{31a} and RGO/CNT/MnO₂ (319 F g⁻¹ at 0.5 Ag⁻¹).^{31b} Based on the literature, the enhanced specific capacitance of the ternary hybrid electrodes was mainly due to the synergistic effect between the each individual components.^{31c}

In this line, the present work explores the synthesis of a ternary composite by an eco-friendly approach and exploits it as a hybrid supercapacitor material. The ternary hybrid composite is prepared by three steps; (i) the hydrothermal synthesis of MnFe_2O_4 , (ii) the environmentally friendly approach is used for preparing graphene using NaOH as the reducing agent and (iii) an *in situ* chemical oxidative polymerization method for preparing the ternary hybrid composite. In order to optimize the electrochemical performance of the hybrid electrode, we have analysed the effect of graphene loading and the PANI concentration. Here, the loading of graphene (20 wt%) and PANI (5 wt%) is optimized with a suitable proportion of components in the composite for hybrid supercapacitor applications. In this, we realized that when the graphene loading was increased beyond 20%, the reversibility of the electrode became decreased. Subsequently, the increase of the diffusive path length is notified with the increase of the PANI concentration. The hybrid supercapacitor was constructed using a hybrid (MGP) electrode and activated carbon as the negative and positive electrode, respectively. To the best of our knowledge, there is no report available for an MGP||AC hybrid supercapacitor. This hybrid supercapacitor is possessed with an enhanced specific capacitance (51.87 F g⁻¹ at 5 mV s⁻¹), energy density (10.25 W h kg⁻¹ at 10 mA cm⁻²) and a superior cyclic stability (100%). In the future, it may offer a new path for the construction of a high energy and specific capacitance hybrid supercapacitor.

2. Experimental methods and materials

2.1 The preparation of pristine MnFe_2O_4

MnFe_2O_4 nanocubes were prepared by the hydrothermal method using stoichiometric amounts of $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and NaOH as the starting compounds. Initially, the cationic precursors were dissolved in double distilled water individually and mixed together under stirring. Finally, 5 ml of an NaOH solution was added drop by drop into the above mixture under vigorous stirring. Subsequently, the mixed precursors were transferred into a 75 ml Teflon lined autoclave. The reactions were carried out at 180 °C for different time periods including 1 h (M-1), 6 h (M-2) and 12 h (M-3). Then, the autoclave was allowed to cool down naturally. Finally the resultant product was washed with distilled water and ethanol several times and dried at 100 °C overnight.

2.2 The preparation of the binary composites

Graphene oxide was prepared using a modified Hummers method,^{32,33} where 2 g of graphite was added into 60 ml conc. H_2SO_4 and stirred for 30 minutes. Next, 6 g of KMnO_4 was added pinch by pinch under stirring in order to avoid overheating and explosion and then it was left for another 150 minutes under stirring. Subsequently, 90 ml of distilled water was added drop by drop into the above mixture. After 15 minutes, 150 ml of distilled water and 5% of H_2O_2 were added to terminate the reaction. Next, 5% of HCl was added to remove the metal oxide impurities. The obtained dark brown colour graphite oxide was washed several times in distilled water until the pH reached 7. It was ultrasonicated for 60 min to convert graphite oxide into graphene oxide. The obtained graphene oxide was further washed with distilled water until it reached pH = 7 and dried at 80 °C for 24 h.

Subsequently, the graphene oxide is reduced to graphene *via* a green route of the hydrothermal method. 50 mg of graphene oxide is dispersed in the concentration of 1 g l⁻¹ water and stirred for 60 min. Next, 8 M NaOH solutions are added drop wise into the above dispersion to reach pH = 12. During the addition of NaOH, the colour of the solution changed from dark brown to black which indicates the partial reduction of graphene oxide. The mixed solution is now transferred to a 75 ml Teflon lined autoclave and kept in a muffle furnace at 180 °C for 12 h. Finally, the precipitate was collected and washed repeatedly. The hydrothermal method reduction has wider advantages than the conventional hydrazine reduction method because hydrazine is highly poisons. The benefits of the hydrothermal reduction are a simple setup, nontoxic reducing elements and defect less graphene.^{34,35a} A similar type of procedure has been adopted by Fan *et al.*,^{35a} where NaOH and KOH were used as the reducing agents for reducing graphene oxide into graphene. But the exact mechanism for the reduction process is still unclear. However, it is strongly believed that the reverse process of the oxidation of graphite into graphene oxide in a strong acidic medium and *vice versa* in alkali medium.

The MnFe_2O_4 nanocube decorated flexible graphene composite was prepared using a hydrothermal method with various weight (wt%) loadings of graphene such as 5 (MG-5), 10 (MG-10), 20 (MG-20) and 30 wt% (MG-30). Initially, 5 wt% of graphene was dispersed in 45 ml of distilled water under stirring. Stoichiometric amounts of $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ and $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ were added into the solution. Finally, 5 ml of an NaOH solution is added drop by drop into the above mixture under vigorous stirring and is continued for few hours. The solution is transferred to a 75 ml Teflon lined autoclave and placed in the furnace for 12 h at 180 °C. The obtained final product was washed with distilled water and ethanol several times and dried at 100 °C overnight. The same procedure was repeated for the preparation of 10, 20 and 30 wt% loading of graphene.

2.3. The preparation of the ternary composite

The ternary composite of MG wrapped PANI (MGP) is prepared using *in situ* chemical oxidative polymerisation. In this synthesis the desired composite was dispersed in a 40 ml mixture solution of aniline (0.15 ml) and H_2SO_4 (4.26 ml) under stirring. The oxidant of ammonium peroxydisulfate (APS) (1.426 g) was dissolved in 50 ml of distilled water and was added drop by drop to the above mixture under stirring. The colour of the mixture changes to green which indicates the presence of PANI in this composite. The stirring was continued for 2 h. The final product was washed and dried and it is named MGP-5 (5 wt%). The same procedure is repeated for the preparation of MGP-10 (10 wt%) and MGP-15 (15 wt%).

2.4 Characterizations techniques

The phase purity and crystallinity of the samples were analyzed using a Bruker D8 Advance with $\text{Cu K}\alpha$ radiation XRD instrument. The presence of functional groups in the samples is analyzed using a Bruker Tensor27 FT-IR instrument. The Raman analysis of our samples is done with the instrument Renishaw Invia Raman Microscope (He-Ne Laser 633 nm, 18 mW). The nano structure and morphological features of the samples were analysed using FE-SEM (Quanta FEG 250) and HRTEM (JEOL JEM 2100). The Bio-logic SP 150 electrochemical workstation is used for analyzing the electrochemical properties of the samples.

2.5 Electrode preparation

The negative electrode was prepared by the following procedure. The as-prepared materials, carbon black and polyvinylidene difluoride (PVDF) were mixed in a mass ratio of 80 : 15 : 5 and dispersed in 0.3 ml *N*-methyl 2-pyrrolidone (NMP) to produce a homogeneous paste. PVDF and carbon black were used as the binder and conductive agent, respectively. Then the resulting mixture of 10 μl was coated onto the stainless steel (SS) substrate (1 cm^2). Finally, the prepared electrodes were dried at 60 °C overnight in a vacuum oven. The resulting active mass of the coated electrode is 1 mg. The electrochemical properties of the electrode were explored under a three-electrode system. The active material coated SS,

platinum wire and Ag/AgCl were used as the working, counter and reference electrodes, respectively. The electrochemical measurements were obtained from cyclic voltammograms (CV), galvanostatic charge/discharge curves (GCDs) and electrochemical impedance spectroscopy (EIS) using a Bio-Logic SP 150 electrochemical workstation.

For a hybrid supercapacitor, the mass balancing between the electrodes is important and it is given in the ESI.† The same procedure has been repeated for the preparation of the positive electrode. Here, 16 μl of slurry was coated on the SS and the active mass is 1.6 mg. The positive and negative electrodes were separated by a polypropylene sheet. Before undergoing electrochemical investigation, the electrodes and separator were immersed in a 1 M NaCl electrolyte for 1 h. The detailed schematic representation for the electrode fabrication and the charge storage mechanism was given in the ESI (Fig. S1†). The fabricated devices electrochemical properties were investigated using CVs, GCDs and EIS in 0 to 1.6 V.

3. Results and discussions

3.1 Structural and morphological analysis

The XRD patterns were used to identify the phase formation of the individual constituents as well as the composites. Fig. 1 (a–b) show the XRD pattern of MnFe_2O_4 prepared at 1 (M-1) and 6 h (M-2), respectively. The XRD pattern of M-1 elucidates the presence of an Fe_2O_3 impurity phase, which may be deficient of reaction time. Furthermore, on increasing the reaction time to 6 h, the product displays the single phase of MnFe_2O_4 with negligible deviation ($\pm 0.01 \text{ \AA}$) in the *d*-spacing values. While further increasing the reaction time to 12 h (Fig. 1(c)), the phase pure with an exact matching of *d*-space was obtained for MnFe_2O_4 . The observed *d*-spacing values are

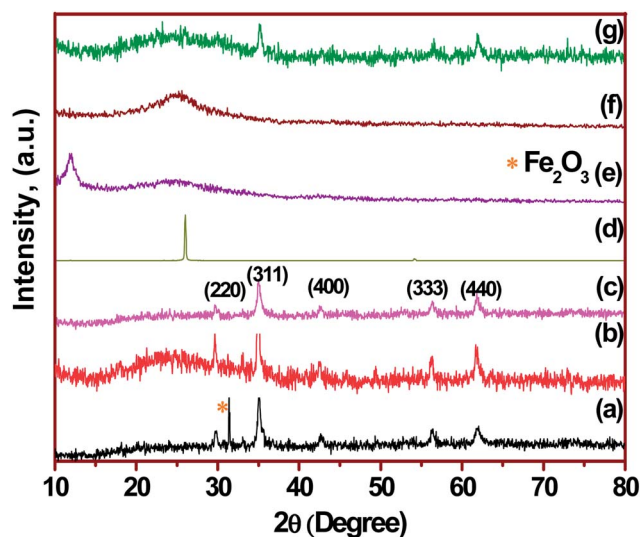


Fig. 1 The XRD patterns of the hydrothermal synthesis MnFe_2O_4 nanocube (a) M-1, (b) M-2, and (c) M-3, (d) pure graphite, (e) graphene oxide, (f) green route reduced graphene, and (g) MG-20.

3.0094, 2.5613, 2.1214, 1.6330 and 1.5003 Å for the corresponding planar diffractions (220), (311), (400), (333) and (440), respectively. It is exactly matched with the JCPDS data (no. 74-2403) with a deviation of ± 0.001 Å. It documents that MnFe_2O_4 has a cubic structure. The calculated lattice constant is $a = 8.4930$ Å which is in good agreement with the reported values.^{12–15} Fig. 1(d) shows the XRD pattern of the pure graphite powder. It shows a single sharp and high intensity peak at 26° corresponding to the d -spacing of 3.42 Å which confirms the disordered nature of graphite. Fig. 1(e) shows a prominent peak at 12° and the disappearance of the peak at 26° which enumerates the conversion of graphite to graphene oxide. Similarly, the d -spacing value is increased to 7.39 from 3.42 Å. According to Lian *et al.*³⁶ the increase of the d -spacing of graphene oxide is due to the intercalated water molecules and the carboxyl and epoxy functional groups in the graphene sheet.³⁶ Fig. 1(f) shows the XRD pattern of graphene. The observed broad peak at 20° to 30° and its corresponding d -spacing value is 3.62 Å.^{32,33,36} In addition, the peak obtained at 12° disappears, which reveals the complete reduction of graphene oxide to graphene. The binary composite of MG-20 formation is confirmed *via* the observed broad peak at 20 to 30° and small sharp peak at 32.5° which confirms the formation of an MnFe_2O_4 nanocube decorated flexible graphene composite (Fig. 1(g)). There are no other graphite and graphene oxide peaks observed in the composite which suggests that during the hydrothermal reaction, the precursor solution is impregnated into the graphene layers, where the nucleation occurs and subsequently the particles are grown inside the layer. Therefore the formed particle acts as a spacer and prevents the reattachment of the graphene layers which in turn increases the interfacing between the graphene layers.^{28,29,35b,37}

Subsequently, FT-IR analysis is employed to identify the presence of functional groups and the phase formation of graphene oxide, graphene, M-3, MG-20 and MGP-5. The FT-IR spectra of graphene oxide and graphene are given in the ESI (Fig. S2†). From Fig. 2(a), two characteristic peaks of MnFe_2O_4 are observed at 481 and 571 cm^{-1} .³⁸ There are no other metal oxide peaks observed which evidenced the single phase formation of MnFe_2O_4 nanocubes. Fig. 2(b) shows the FT-IR spectrum of MG-20. In this, three main characteristic peaks of graphene and MnFe_2O_4 are observed at 1565, 566 and 476 cm^{-1} .³⁶ This reveals the formation of a binary hybrid composite of MnFe_2O_4 /graphene. Furthermore, the disappearance of the peaks at 1735 cm^{-1} , 1125 cm^{-1} and 1053 cm^{-1} for the carboxyl and epoxy functional groups suggests that graphene oxide is completely reduced to graphene and its reattachment is prevented by the decoration with the MnFe_2O_4 nanocubes. Fig. 2(c) shows the FT-IR spectrum of MGP-5. It shows extra characteristic bands at 1398, 1507 and 1580 cm^{-1} , corresponding to the plane bending vibration of C–H and the C=C stretching vibration of the benzenoid and quinoid rings of PANI, respectively.^{39,40} Furthermore, the characteristic peaks of MnFe_2O_4 are shifted to a higher wavelength region (571 to 556 cm^{-1} and 481 to 472 cm^{-1}). This reveals the formation of a ternary hybrid MGP

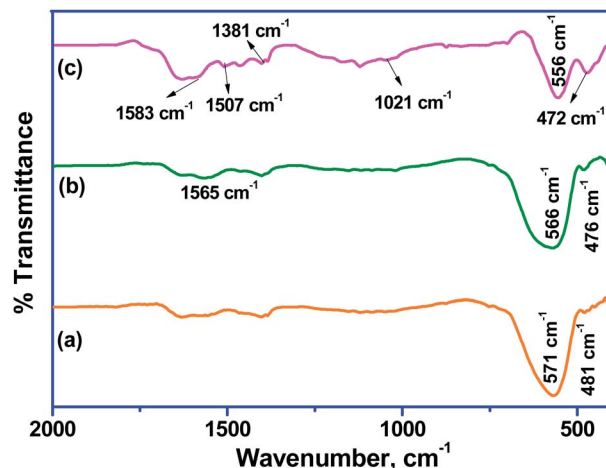


Fig. 2 The FT-IR spectra of the (a) M-3, (b) MG-20 and (c) MGP-5 samples.

composite. Similarly, the absorption band observed at 1021 cm^{-1} reveals the presence of defects in the graphene. However, the defect peak is not observed in MG-20. Hence, the defect may be due to the incorporation of PANI.

Raman spectroscopy is employed to identify the presence of defects as well as the phase formation. The representative Raman spectra of graphene and MGP-5 are given in Fig. 3(a and b). The Raman spectrum of graphene (Fig. 3(a)) shows two prominent peaks at 1339 and 1588 cm^{-1} , which correspond to the D and G modes. G denotes phonon emission from the first order zone midpoint and is related to the sp^2 carbon vibration in the graphene domains. In addition, D denotes the presence of defects in the graphene which is a combination of edge and structural changes. It can be seen that the intensity of the D band is small compared to the G band which indicates the presence of a lower number of defects in the prepared graphene. In addition, it witnessed that graphene contains few layers of graphene sheets. The intensity ratio is 0.61 which is less than the reported values.^{32,33,41} Furthermore, a planar

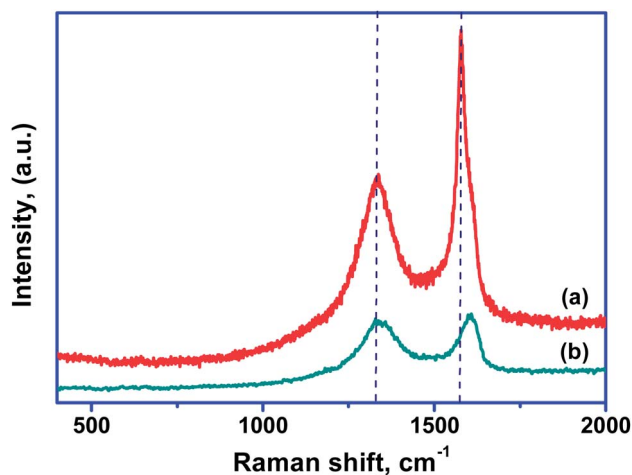


Fig. 3 The Raman spectra of (a) graphene and (b) MGP-5.

domain size is calculated which is equal to 7.21 nm.³³ This large domain size graphene may act as a substrate to anchor the large number of MnFe_2O_4 nanocubes and effectively enhance the electrochemical utilization and active sites of the MnFe_2O_4 nanocubes. Fig. 3(b) shows the Raman spectrum of the ternary composite (MGP-5). The peak broadening of D and G is observed when compared with pristine graphene. It may be due to the presence of functional groups of PANI.^{23,24} Furthermore, the calculated intensity peak ratio is 0.91 which is higher than that of the pure graphene. It elucidates that the presence of PANI inhibits the reattachment of the graphene layers and also partially increases the imperfections in graphene which in turn enhances the accessibility of the ions into the electrode. In addition, the calculated planar domain size is (4.85 nm) comparable with graphene. Hence, it concludes that the presence of PANI substantially decreases the domain size, and increases the defects in graphene and the intensity ratio (I_D/I_G).⁴¹

Fig. 4(a) shows the TEM image of graphene. It reveals that the graphene consists of a few layers. The observed ensnared and waves on the surface evidenced the mechanical stability of the graphene. Furthermore the observed transparent graphene sheets directly evidences the formation of single layer graphene. Furthermore, the dot patterns from the selected area electron diffraction pattern (Fig. 4(b)) infer the crystalline nature of graphene. Fig. 4(c) shows the FE-SEM image of MGP-5. The MnFe_2O_4 nanocubes are uniformly distributed on the flexible graphene sheets and they are covered by PANI. Similarly, the size of the individual nanocubes is in the range between 75 and 100 nm, which is measured using a scion image. On the other hand, MGP is shown to be more transparent than graphene. It infers that the graphene is reduced to single layered graphene by the impregnation of MnFe_2O_4

and PANI. In the TEM image (Fig. 4(d)), it can be seen that the PANI films are wrapped on the MnFe_2O_4 nanocubes and graphene. Here, graphene and PANI act as a conductive network and enhances the electrochemical utilization of ions from the electrolyte into the electrode for the reduction of the diffusive path length of the ions. In addition, the flexibility of graphene sheet will enhance the electrochemical stability during charge/discharge cycling.

Based on the previous results, the formation mechanism (Fig. 5) of MGP can be explained as follows; (i) the adsorption of metal ions on the surface of graphene, (ii) nucleation and growth, and (iii) interaction.³⁷ In this process, the starting cationic precursors are dissolved in water which provides Mn^{2+} and Fe^{3+} ions. The small Mn^{2+} and Fe^{3+} ions undergo coalescence, the joining of small particles, in the presence of NaOH. Furthermore, the particles are attracted towards graphene due to its high surface energy. Hereby, the formed small particles act as nucleation seeds to grow MnFe_2O_4 at the interface between the solution and graphene *via* the van der Waals force of attraction and magnetic dipolar interaction between the initially formed particles. Finally, PANI is wrapped over the MnFe_2O_4 nanocube decorated flexible graphene *via* π - π interactions and hydrogen interactions between PANI and graphene and PANI and MnFe_2O_4 using chemical oxidative polymerisation.

3.2 The electrochemical properties of the single electrode

The electrochemical properties of the pristine MnFe_2O_4 and the binary composite. To identify the appropriate electrolyte of MnFe_2O_4 , the CV was measured at 10 mV s^{-1} in three different electrolytes of 1 M NaCl, 1 M KOH and 1 M H_2SO_4 and is given in Fig. 6. It is well known that the current area covered in the CV curve is directly proportional to the amount of charge stored in the system.^{42,43a} Hence, the MnFe_2O_4 exhibits the specific capacitance of 45, 34 and 9 F g^{-1} at 5 mV s^{-1} in 1 M NaCl, 1 M KOH and 1 M H_2SO_4 electrolytes, respectively. From the Fig. 6, it can be seen that the MnFe_2O_4 nanocubes are completely utilized in aqueous 1 M NaCl compared to the alkali 1 M KOH

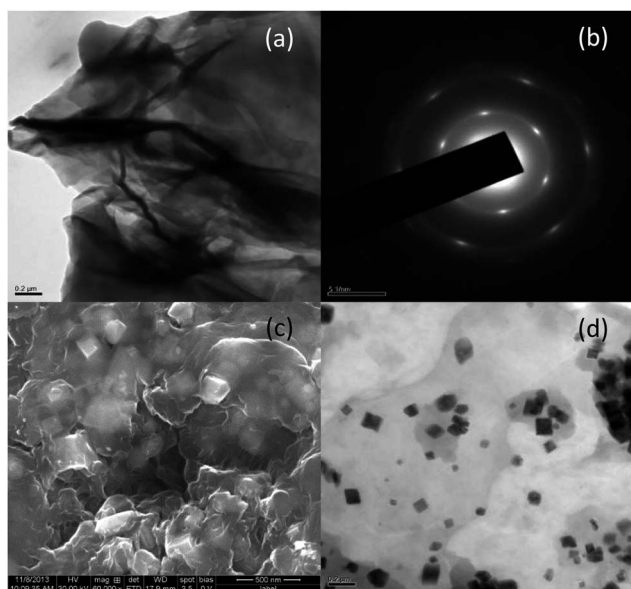


Fig. 4 (a and b) A TEM image of graphene and its corresponding SAED pattern, and (c and d) FE-SEM and TEM images of MGP-5.

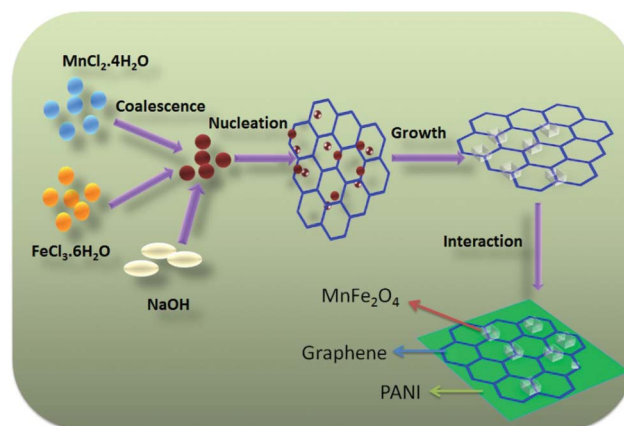


Fig. 5 A plausible formation mechanism for the MnFe_2O_4 nanocube decorated flexible graphene with wrapping of PANI.

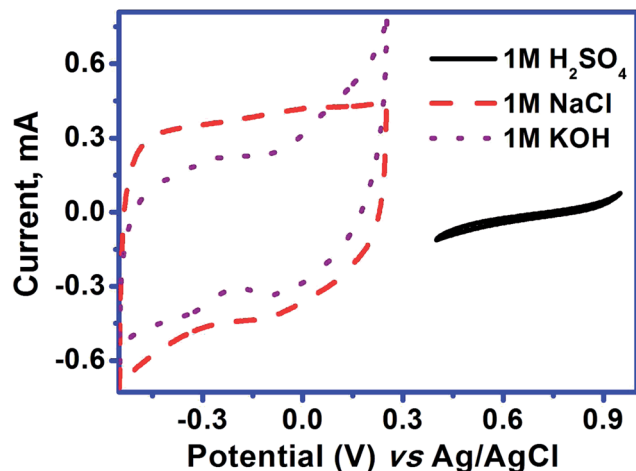


Fig. 6 The CV curves of M-3 in the 1 M NaCl, 1 M KOH and 1 M H₂SO₄ electrolytes at 10 mV s⁻¹.

and acidic 1 M H₂SO₄ electrolyte. Based on this, 1 M NaCl is recognized as the appropriate electrolyte for the subsequent exploration. However, the experimental specific capacitance is lower than that of the theoretical specific capacitance (522 F g⁻¹), which may be due to less contribution from the active sites (0.1183) and a lower specific surface area.^{42,43a}

In order to further augment the specific capacitance and energy density of the electrode, the MnFe₂O₄ nanocubes are combined with the different weight loadings of graphene. Fig. 7(a) shows the CV curves of all the samples at 100 mV s⁻¹. It demonstrates that while increasing the graphene loading from 5% to 20%, the rectangularity and the integrated area of the CV curve increases. Hence, it documents the augmentation of the electrochemical reversibility and electrochemical utilization which mends the specific capacitance of the electrode. However, the CV behaviour of MG-30 is deviated from rectangularity when compared to the other loadings (Fig. 7(a)) which

indicates that the reversibility of the electrode decreases at higher loading of graphene.

Actually, at a higher loading, the graphene undergoes reattachment due to strong van der Waals forces of interaction that reduce the ionic accessibility and increase the electrode resistance.^{43b,43c} It is evident that at the reverse switching potential of the CV curve (MG-30), the current is gradually decreased to reach the potential axis when compared with the other composites.^{43d} Hence, the higher loading of graphene (MG-30) is not suitable for supercapacitor applications due to the low electrochemical reversibility and the large internal resistance at a high scan rate. Therefore, MG-20 is recognized as the optimized electrode for further studies based on an improved reversibility and low internal resistance. The CV curves of the samples at different scan rates are given in Fig. S3† and they illustrate that the integrated area in the CV curve is increased with the increasing scan rate.^{42,43a} In addition the rectangularity of the CV curves is preserved even at high scan rates except for MG-30 which reveals the good reversibility of the electrode.

Fig. 7(b) shows the variation of the specific capacitance as a function of scan rates from 5 to 100 mV s⁻¹ in 1 M NaCl. At a low scan rate, the sample MG-30 exhibits a higher specific capacitance than other samples. Even though it gives a higher specific capacitance, it gets diminished while increasing the scan rate more than any other composite. In order to further evidence the above results, the observed linear fitting of the discharge current (Fig. 7(c)) largely deviates from the linear fit with that of the scan rate for MG-30. This concludes that MG-30 has a lower electrochemical reversibility and a high electrode resistance. The decrease of the specific capacitance with the increasing scan rate is further substantiated by calculating the area occupied by hydrated Na⁺ ions (S) on the surface of the electrodes with respect to scan rate (Fig. 7d). It revealed that (i) at lower scan rates, the ions from the electrolyte have the ability to diffuse into the inner surface of the electrode and lodge more in the active area in the electrode due to a high accessible time, and (ii) at higher scan rates, the ions lodge in only a limited area in the electrode due to their short time access of ions from electrolyte. The specific capacitance is high at low scan rate and *vice versa*.^{6,42,43a} The decreasing order of the area occupied by the Na⁺ ions into electrode can be represented as MG-20 (136 m² g⁻¹) > MG-30 (117 m² g⁻¹) > MG-10 (94 m² g⁻¹) > MG-5 (64 m² g⁻¹) > M-3 (60 m² g⁻¹) at a scan rate of 100 mV s⁻¹. Furthermore, GCDs analysis is carried out to substantiate the CV results. The calculated specific capacitance is 66, 91, 136, 189 and 309 F g⁻¹ for M-3, MG-5, MG-10, MG-20 and MG-30, respectively.⁴⁴

However, the experimental specific capacitance is lower than the theoretical specific capacitance (528 F g⁻¹) due to the sinking of the active site (0.27607). On the other hand, the energy density is enhanced on increasing the graphene loading such as 2.5, 3.9, 4.55, 8.51 and 12.03 W h kg⁻¹ for M-3, MG-5, MG-10, MG-20 and MG-30, respectively. Finally it is clear that the energy density of the binary composite is 3.5-fold higher than that of pristine MnFe₂O₄. In order to substantiate the optimization of the graphene loading in terms of the charge transfer mechanism at the electrode/electrolyte interface, the

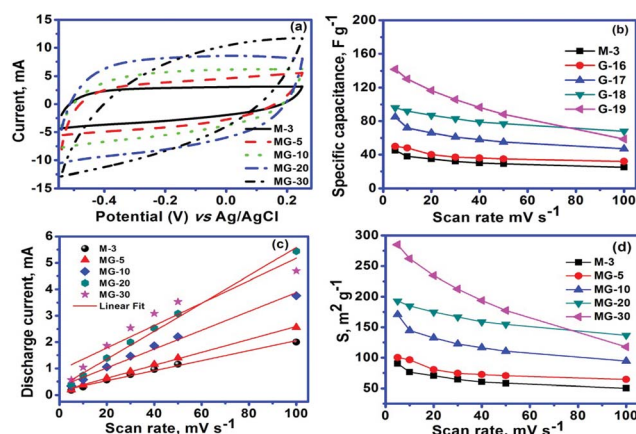


Fig. 7 A comparison of the cyclic voltammograms of MnFe₂O₄ and the binary composite (a) at a scan rate of 100 mV s⁻¹, (b–d) plots for the specific capacitance, linear fit of discharge current and area occupation of the hydrated ions *versus* different scan rates.

electrochemical impedance analysis (EIS) is carried within the frequency range of 0.1 MHz to 10 mHz in the perturbation potential of 10 mV. It is well known that the supercapacitor behaves as a resistor at the high frequency region and as a pure capacitor at the low frequency region.⁴⁵ Generally, the frequency range is divided into three different regions; (i) high frequency, (ii) middle frequency, and (iii) low frequency. At the high frequency region, the intersection of the semicircle is explained based on the equivalent series resistance (ESR), it is the combination of the internal resistance of the electrode material, the interfacial resistance at the active material and substrate and the solution resistance.⁴⁶ The EIS plot is fitted with the equivalent circuit and is given in Fig. S4.† The measured ESR values (Fig. 8(a)) are 3, 3.1, 3.2, 3.9 and 4 Ω which correspond to the composites M-3, MG-5, MG-10, MG-20 and MG-30, respectively. It explains that there is no significant difference in ESR due to the loading of graphene. Similarly, at the mid frequency region, a single semicircle is observed. This indicates the electrochemical charge transfer resistance at the electrode and electrolyte interface. It demonstrates that the interfacial charge transfer resistance is decreased with increasing the graphene loading up to 20 wt% (MG-20) after that it gets increased for 30 wt% (MG-30). The decreasing order of the R_{ct} values is M-3 (18.98 Ω), MG-30 (8.342 Ω), MG-5 (5.826 Ω), MG-10 (3.779 Ω) and MG-20 (1.895 Ω). It evidenced that the sample MG-30 having less electrochemical activity which may be due to the reattachment of graphene at a higher loading. Other fitted parameters are tabulated in Table S1.† Subsequently, at the low frequency region, a single tail is observed which indicates the capacitive behavior of the electrode. In general, for an ideal capacitor, the low frequency tail is parallel to the imaginary axis. On the other hand, the deviation of the tail from the imaginary axis (approximately 45°) demonstrates the pseudocapacitive behavior of the material. In the present case, the tail is approaching the imaginary axis when increasing graphene loading. It documents that the ideal capacitive behaviour gets improved by the addition of graphene. Fig. 8(b) elaborates that

at the lower frequency region, the total impedance of the composite is decreased with increasing the loading of graphene up to 20% (MG-20) after which a sudden increase of impedance is notified. The phase angle approaches 90° (Fig. 8(c)) with increasing the graphene loading which indicates the more contribution of EDLC.^{47,48} It also says that the charge storage mechanism is based on both EDLC and pseudocapacitive behavior. Hence, the graphene acts as an anchor to hold the MnFe_2O_4 particles which enhances the accessibility of the ions from the electrolyte. Finally, it reduces the charge transfer resistance and mends the specific capacitance and reversibility of the electrode. In addition, at the high frequency region, a single peak is observed and is denoted by ϕ . It can be seen that ϕ is shifted to a higher frequency region while increasing the graphene loading up to MG-20, which envisages the reduction of the diffusive resistance of ions into the electrode.⁴⁸ This reduced diffusive resistance, enhances the electrochemical utilization of the ions into the electrode that leads to higher specific capacitance. In contrast, MG-30 shows the almost similar ϕ value of M-3, *i.e.*, the composite has a high concentration of graphene which leads to a higher diffusive resistance. Furthermore, the admittance plot (Fig. 8(d)) demonstrates that the knee frequency is increased with graphene loading up to 20 wt% after which the knee frequency is shifted to a lower frequency region. The observed knee frequencies are 17.52, 125.2, 406.69, 1963.14 and 26.97 Hz. This shows that the relaxation time is decreased with the graphene loading. It means that the resistive behavior of the electrode is quickly changed to a capacitive behavior when increasing graphene loading up to 20%. The calculated relaxation times are 57, 7, 2, 0.5 and 38 ms for M-3, MG-5, MG-10, MG-20 and MG-30, respectively.⁴⁹ Based on the above results, the optimized sample for further analysis is MG-20.

Electrochemical properties of ternary composites. Subsequently, to further improve the specific capacitance of the electrode, the optimized MG-20 is combined with PANI in different weight percentages such as 5, 10 and 15 wt%. The CV curve of MGP-5, MGP-10 and MGP-15 (Fig. 9(a)) reveals that the current area is enhanced compare with that of a binary composite. In other words, the area under the CV curve is increased on increasing the PANI concentration. In addition, the observed redox peaks evidenced the presence of PANI in the composite and its pseudocapacitive contribution further supports to enhance the specific capacitance.⁵⁰ This redox peak is due to the transition from the leucoemeraldine form to the emeraldine form. Hence, the ternary composite electrode stores the charge both EDLC and pseudocapacitive mechanisms. The pseudocapacitive contribution is dominant in MGP-10 and MGP-15 compared to MGP-5 which is clearly evidenced from the CV curves. The calculated specific capacitance is 244, 247 and 240 F g^{-1} at 5 mV s^{-1} for MGP-5, MGP-10 and MGP-15, respectively. It is higher or comparable with previously reported values.^{25,30a} Furthermore, effect of the scan rate on the electrochemical properties of MGP-5, MGP-10 and MGP-15 is investigated (Fig. S5†). It evidenced that the rectangularity of MGP-10 and MGP-15 is completely deviated when compared to MGP-5 which demonstrates the

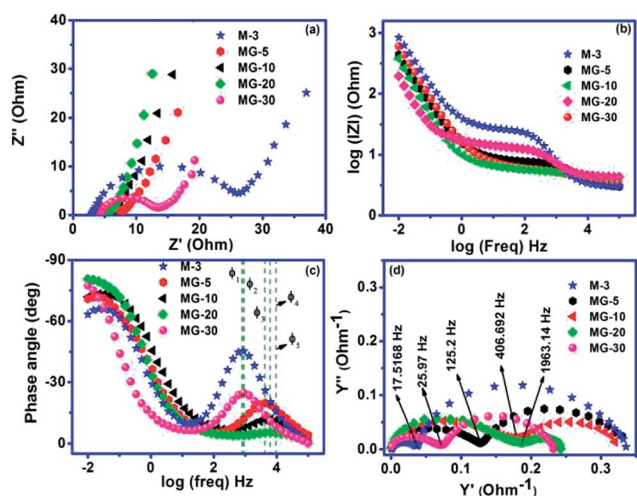


Fig. 8 (a) The Nyquist plot, (b and c) Bode plot and (d) admittance plot of MnFe_2O_4 and binary composites in 1 M NaCl electrolyte.

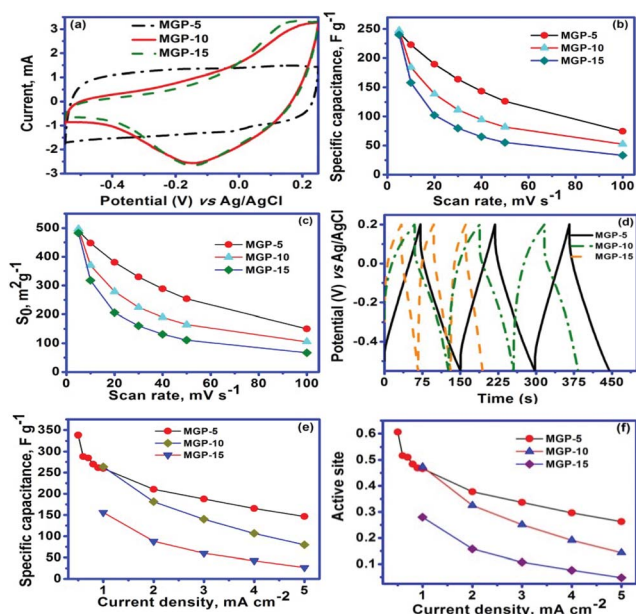


Fig. 9 The CV curves of (a) the ternary composite at 5 mV s^{-1} , (b and c) the plot for the specific capacitance and area occupation of the hydrated ions *versus* different scan rates, (d) the GCDs curves of the ternary composites at 2 mA cm^{-2} and (e and f) plot for specific capacitance and active site *versus* at different current densities.

increase of the internal resistance of the electrode. It may be due to the thick wrapping of PANI which limits the diffusive path length. Hence, it reveals that a low concentration of PANI is favourable for the ternary composite.^{31a,51a} Nevertheless, at a low scan rate, the electrode materials exhibit almost similar specific capacitance.^{25,51b} As evidenced from the CV curve (Fig. 9(a)), it is strongly believed that at a low concentration of PANI, the capacitance contribution is mainly due to the EDLC mechanism. On the other hand, while increasing the PANI concentration, the redox reaction is predominant. Hence, the charge storage mechanism played an important role for the occurrence of specific capacitance.²⁵

Fig. 9 (b, c) shows the calculated specific capacitance and the area occupied by hydrated Na^+ ions on the electrode surface with respect to scan rate. It evidence that at a low scan rate, all the samples are almost equally utilized and the areas covered by the ions are almost equal because the ions from the electrolyte have a huge time to utilize both the inner and outer surface of the electrode. But at a high scan rate, the area occupied by ions from the electrolyte for MGP-10 and MGP-15 is drastically decreased compared to MGP-5 which may be due to the reduction of the diffusive path length. The thick wrapping of PANI restricts the ions from the electrolyte in completely utilizing the electrode and also the ions from the electrolyte have a limited accessible time into the electrode. It leads to a decrease in the specific capacitance of the electrode. Hence, the optimized sample for further investigation is MGP-5 based on the reversibility and enhanced specific capacitance.

It is well known that the total specific capacitance of the electrode is the combination of the specific capacitance contributed by the inner (C_{in}) and outer (C_{out}) surface of the electrode. It can be calculated using a Trasatti plot (Fig. S6†). According to Trasatti, the total specific capacitance is the sum of the specific capacitance provided by the inner and outer surface of the electrode. It can be expressed as^{51c}

$$C_{\text{total}} = C_{\text{in}} + C_{\text{out}} \text{ F g}^{-1} \quad (1)$$

The specific capacitance contribution from the inner and outer electrode depends on the scan rate (v). The plot between C_{total} *vs.* $v^{1/2}$ exhibits the total specific capacitance of the y -axis intercepted at $v = 0$. The diffusion of the ions into the electrode is uncontrolled. Subsequently, the plot between C_{total} *vs.* $v^{-1/2}$ represents the specific capacitance contribution from the outer surface of the electrode at $v = \infty$. The diffusion of the ions into the electrode is controlled. As seen from Fig. S7† the total specific capacitance of MGP is higher than M-3. Furthermore, the specific capacitance which is provided by the inner surface of MGP is much higher than M-3 and MGs. This result suggests that the presence of PANI and graphene act as a conductive network which reduce the electron transport path length and the diffusion length of the ions into the electrode. This leads to improve the inner active site of MGP and exhibits more amount of charge storage in the inner electrode. The total specific capacitance exhibited by the electrode is 46.76, 53.13, 88.63, 103.31, 164.55, 259.49, 262.52 and 240 F g^{-1} , for M-3, MG-5, MG-10, MG-20, MG-30, MGP-5 and MGP-10, respectively. This impressively well supports the previous results.

For more evidence, the GCDs analysis is carried out for the ternary composite electrode at different current densities such as 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 2.0, 3.0, 4.0 and 5.0 mA cm^{-2} . Fig. 9(d) shows the comparative GCDs curves for MGP-5, MGP-10 and MGP-15 at a current density of 2 mA cm^{-2} , in which the IR drop is observed at the beginning of the discharge time. The IR drop value tends to increase with the increasing PANI concentration. This is due to the lower electronic conductivity of PANI in 1 M NaCl.^{51a} The charge/discharge curve is largely deviated from the linearity for MGP-10 and MGP-15. This is evidence that the charge storage mechanism in the ternary composite is mainly contributed by both EDLC and pseudocapacitive behavior. Furthermore, the specific capacitance values for MGP-5, MGP-10 and MGP-15 are almost equal which is the same as the CV results. The maximum attained specific capacitance is 338 F g^{-1} at a current density of 0.5 mA cm^{-2} . The obtained specific capacitance is 10 times higher than M-3. The obtained specific capacitance is higher than the reported values.^{28,30b,31b} However, the experimental specific capacitance is lower than the theoretical specific capacitance (601 F g^{-1}), due to the less involvement of the active site (0.6059). Subsequently, the specific capacitance of the (Fig. 9(e)) electrode is decreased with the increasing current densities. This can be explained based on the ion interaction time between the electrode and ions from the electrolyte, and the active sites (Fig. 9(f)). At a low current density, the ion accessibility into the electrode is high

due to their enormous access time which leads to enhance the active sites of MnFe_2O_4 . Furthermore, this leads to a high specific capacitance at low current densities.^{42–44} Afterwards, at a higher current density, the ions have the ability to interact only with the outer surface of the electrode due to their limited access time which facilitates only limited active sites. Hence, at a high current density, the electrode possesses a lower specific capacitance. MGP-5 exhibits the maximum energy density of 26.40 at a current density of 0.5 mA cm^{-2} . The energy density of MGP-5 is 10.56 times greater than MnFe_2O_4 . In addition, it is greater or comparable to the energy density of graphene/ SnO_2 /PEDOT (22.8 W h kg^{-1}).³⁰

Furthermore, to substantiate the previous results, the EIS analysis was carried out for MGP-5, MGP-10 and MGP-15. MGP-10 and MGP-15 has a larger charge transfer resistance than MGP-5 (Fig. 10(a)) due to the poor electronic conductivity of PANI in 1 M NaCl.^{51a} Furthermore, the low frequency tail of MGP-5 is almost parallel to the imaginary axis. In contrast the tail of MGP-10 and MGP-15 approaches the real axis which may be due to the pseudocapacitive contribution from surface wrapped PANI. It infers that MGP-5 has a lower diffusive resistance than MGP-10 and MGP-15. Subsequently, the calculated R_{ct} values are 23.04 Ω , 32.76 Ω and 35.76 Ω for MGP-5, MGP-10 and MGP-15, respectively. Other fitted parameters are tabulated in Table S1.† In addition the Bode plot (Fig. 10(b and c)) also implies that the resistance of MGP-10 and MGP-15 is higher than MGP-5 and the phase angle closely approaches 90 degrees for MGP-5 compare with MGP-10 and MGP-15. This further substantiates that MGP-5 stores the charge by both an equal

contribution of EDLC and pseudocapacitive behavior. However, the EDL capacitive behavior is dominant in MGP-5. Subsequently, MGP-10 and MGP-15 store the charge mainly based on the pseudocapacitive behavior than EDLC. Subsequently, a single peak (φ) is observed at a high frequency region, which is almost same for MGP-5, MGP-10 and MGP-15 which shows that all the electrodes have almost the same diffusive resistance. Subsequently, the knee frequency (Fig. 10(d)) is shifted to a lower frequency region with increasing PANI loading which means the increasing charge transfer resistance. The observed knee frequency value is 5.5395, 3.6371 and 2.4537 Hz and its corresponding relaxation time is 180, 274 and 407 ms, respectively. Hence, the MGP-5 is quickly changed to a capacitive behaviour when compared to MGP-10 and MGP-15. The diffusive coefficient (D) of ions into the electrode can be found using Randles plot. It is drawn (Fig. 10(e)) between $\omega^{-1/2}$ and Z' . The slope of the plot is represents the Warburg factor (k_w). The relationship between k_w and D can be expressed as;^{51d}

$$D = \left(\frac{RT}{n^2 F^2 A \sqrt{2}} (1/k_w C^*) \right)^{1/2} \quad (2)$$

where, R is gas constant, T is the temperature (K), n is the number of electron transfer, F is Faraday constant, A is the active area in electrode (cm^2), and C^* is the ionic concentration (mol cm^{-3}). Subsequently, the calculated diffusive coefficient is 1.252×10^{-9} , 2.755×10^{-11} and $5.1967 \times 10^{-13} \text{ cm}^2 \text{ s}^{-1}$ for MGP-5, MGP-10 and MGP-15 from the Randles plot (Fig. 10(e)), respectively. This result implies that increase PANI leads to decrease of diffusive coefficient and increase of diffusive resistance. It further substantiates the CV and GCDs results.

Hence, the enhanced electrochemical performance of the ternary composite is due to (i) π - π interaction between PANI and graphene, the hydrogen bond between PANI and MnFe_2O_4 , an enhanced electronic conductivity (ii) the impregnation of the MnFe_2O_4 nanocube into graphene may reduce the diffusion path length of ions, prevent the reattachment of the graphene sheets which facilitates the superior interaction between the electrolyte ions and MnFe_2O_4 , (iii) the graphene sheets act as conductive substrate to reduce the internal resistance, (iv) the introduction of PANI further prevents the reattachment of graphene which leads to enhance the electrochemical utilization, (v) the enhanced area occupation of the hydrated ions and active sites, (vi) the high diffusive coefficient and (vii) the synergistic effect. We believed that it is a new and novel electrode for future supercapacitor applications.^{23–31,40,50,51a}

Electrochemical properties of hybrid supercapacitor cell. For the first time, the hybrid supercapacitor device of MGP-5||AC is constructed and also the working potential window is increased up to 1.6 V in aqueous electrolyte. This enhanced potential window may be due to the best combination of electrodes like MGP-5 and AC for hybrid device. The AC (positive electrode) stores the charge based on the separation of charges at the electrode/electrolyte interface. The hybrid electrode (negative electrode) stores the charge based on the fast redox reaction (Fig. S8†). Fig. 11(a) shows the CV curve of

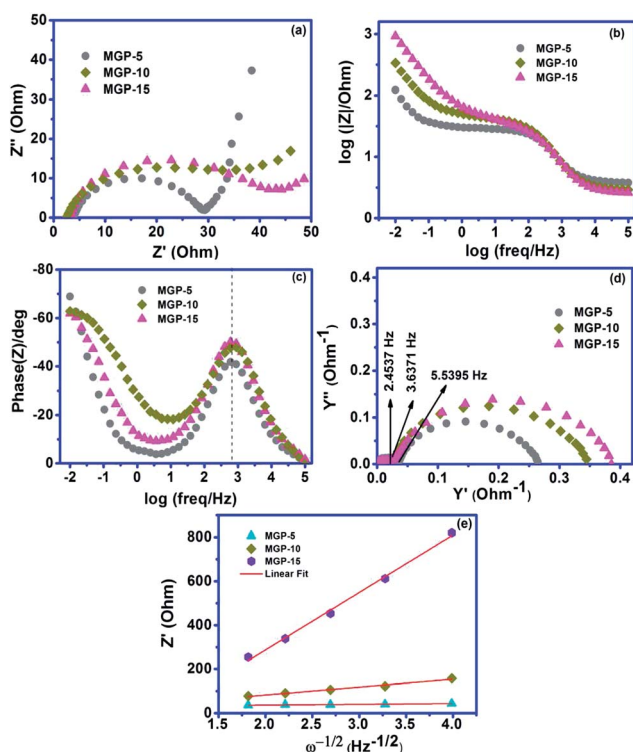


Fig. 10 (a) The Nyquist plot, (b and c) Bode plot, (d) admittance plot and (e) Randles plot for ternary composites in the 1 M NaCl electrolyte.

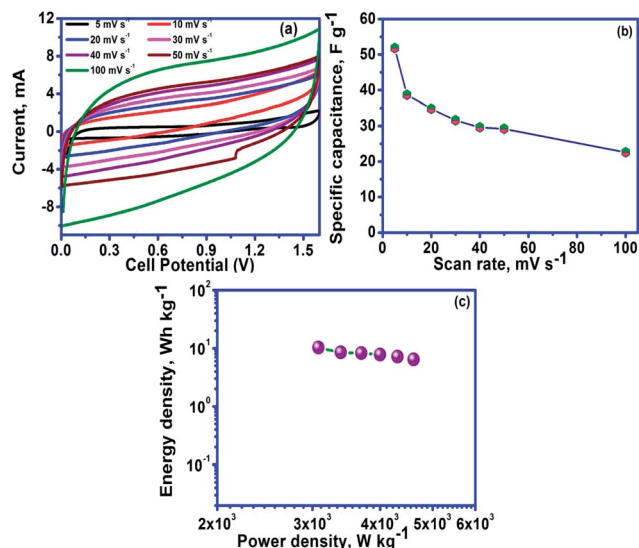


Fig. 11 (a) The CV curve of the hybrid supercapacitor at different scan rates, (b) plots of the specific capacitance versus different scan rates, and (c) the Ragone plot for the MGP-5||AC hybrid supercapacitor.

the device at different scan rates from 5–100 mV s⁻¹. In these CV curves, no O₂ or H₂ evaluations are observed at the two edges of sweeping or switching potential and also the rectangularity of the CV curve is maintained even at a higher scan rate. Hence, this potential window is optimized without damaging the device and has a good electrochemical reversibility. The hybrid system exhibits a specific capacitance of 51.87 F g⁻¹ at 5 mV s⁻¹. Fig. 11(a) shows that the area surrounded by the CV curve is increased with the increasing scan rate. Even at a high scan rate also the rectangularity of the CV curve is maintained. On the other hand, the specific capacitance (Fig. 11(b)) of the device is decreased with the increasing scan rate. The GCD curve displays that the charging and discharging time is almost equal and linear. It indicates the good electrochemical reversibility of the hybrid device. In this, there is no IR drop observed at the beginning of the discharge time even at a higher current density (Fig. S9†). This shows that the device has a low internal resistance. It may reduce the loss of the energy during the charging and discharging of the device. It is well agreed with CV result. The maximum obtained specific capacitance is 37.5 F g⁻¹ at the current density of 10 mA cm⁻². To further explore the electrochemical performance of MGP||AC for supercapacitor, the Ragone plot (Fig. 11(c)) resembles that decrease of energy density with the increasing power density. The hybrid supercapacitor possesses the maximum energy density of 10.25 W h kg⁻¹ at a power density of 3076 W kg⁻¹. It is higher and comparable with the reported values like MnO-RGO||Functionalized carbon (2.6 W h kg⁻¹), MnFe₂O₄||LiMn₂O₄ (10 W h kg⁻¹) and AC||MnO₂ (10 W h kg⁻¹).^{52–54} The enhanced specific capacitance and energy density is attributed to the high ionic conductivity of the aqueous electrolyte and synergistic effect between the electrodes.⁵⁵ For practical applications, the cyclic stability of the device is an important requirement. The cycling stability of the

device is examined over (Fig. 12) 2000 cycles at a current density of 15 mA cm⁻². The corresponding GCDs curve is displayed in the inset of Fig. 12. It shows that the specific capacitance remains unchanged without any decay even after 2000 cycles. Thus the maintained specific capacitance may be due to the effective utilization of MnFe₂O₄ and the high electrochemical reversibility of the system.⁵⁶

The capacitance retention is superior than those obtained from previous reports on PANI/graphene/CNT||activated graphene (91.4% after 5000 cycles) and three dimensional graphene (3DG)/CNT/MnO₂||3DG/MnO₂ (95.3% after 1000 cycles).^{57,58} This indicates that the MGP-5||AC supercapacitor is a suitable electrode for future applications based on its good electrochemical reversibility, its enhanced specific capacitance, and its low internal resistance. In order to know the electrochemical kinetics of the device, the EIS analysis was carried out. It can be seen from Fig. S10(a),† that the Nyquist plot consists of a single semicircle (mid frequency) and a tail (low frequency). The diameter of the semicircle or the charge transfer resistance is almost the same after and before cycling. The *R*_{ct} values are 5.211 and 6.981 Ω for before and after cycling, respectively. Other fitted parameters are tabulated in Table S1.† The Bode plot (Fig. S10(b)†) shows that the resistance of the device before and after cycling is almost equal at the low frequency region which decreases with the increasing frequency. The observed phase angle (Fig. S10(c)†) is approximately 80° and 72° before and after cycling. It is close to 90° from the real axis. Hence, the device has an ideal capacitive nature. Furthermore, at the high frequency region, a single peak is observed after and before cycling. The value of *φ* is almost the same and there is no significant variation. Hence, the device is having approximately same diffusive resistance. The admittance (Fig. S10(d)†) plot shows that the knee frequency is higher before cycling (56.78 Hz) than after cycling (38.42 Hz). The calculated relaxation time is 17 and 26 ms for before cycling and after cycling, respectively. All the above results show that the MGP||AC hybrid supercapacitor device is a promising one for various high power applications.

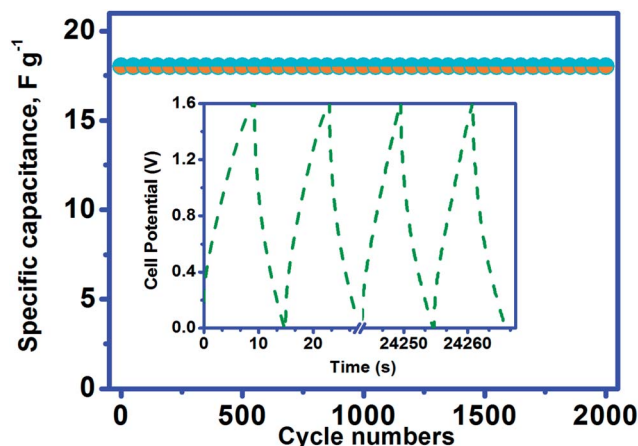


Fig. 12 The specific capacitance versus cycle numbers and the inset is the representative GCDs curve of the initial and final two cycles.

4. Conclusions

In this summary, we have synthesized MnFe_2O_4 nanocube decorated graphene wrapped with PANI as a ternary composite for energy storage applications *via* a green route. The XRD, FT-IR and Raman analysis demonstrates the formation of an MGP composite. Similarly, the FE-SEM and TEM results revealed that the MnFe_2O_4 nanocubes are decorated on flexible graphene and are covered by PANI. The size of the nanocubes is in the range of 75–100 nm. The specific capacitance of the binary electrode is increased with the increasing graphene loading. In addition, increasing PANI not only significantly increases the specific capacitance but also reduces its capacity retention. This shows that PANI has a low electronic conductivity in the 1 M NaCl electrolyte. It gives a higher energy density (approximately 10 times) and a better specific capacitance (approximately 10 times) than pristine MnFe_2O_4 . This may be due to the higher surface area, the high ionic diffusion, the good electronic conductivity, the high active sites, the large area for hydrated ions on the electrode surface, the synergistic effect and the low internal resistance. It proves that MGP is a suitable electrode for the hybrid supercapacitor to work along with a carbon electrode. The specific capacitance is 51.87 F g^{-1} at 5 mV s^{-1} . Furthermore, the hybrid supercapacitor exhibits a good electrochemical cycling stability up to 2000 cycles.

Acknowledgements

The authors would like to thank Department of Science and Technology (DST-SERB), Government of India, for providing financial support (SR/FTP/PS-80/2009) under the “Fast Track Young Scientist” scheme.

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